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Class Start Times, Sleep, and Academic Performance in College: A Path Analysis

Serge V. Onyper, Pamela V. Thacher, Jack W. Gilbert, and Samuel G. Gradess

Department of Psychology, St. Lawrence University, Canton, New York, USA

Path analysis was used to examine the relationship between class start times, sleep, circadian preference, and academic performance in college-aged adults. Consistent with observations in middle and high school students, college students with later class start times slept longer, experienced less daytime sleepiness, and were less likely to miss class. Chronotype was an important moderator of sleep schedules and daytime functioning; those with morning preference went to bed and woke up earlier and functioned better throughout the day. The benefits of taking later classes did not extend to academic performance, however; grades were somewhat lower in students with predominantly late class schedules. Furthermore, students taking later classes were at greater risk for increased alcohol consumption, and among all the factors affecting academic performance, alcohol misuse exerted the strongest effect. Thus, these results indicate that later class start times in college, while allowing for more sleep, also increase the likelihood of alcohol misuse, ultimately impeding academic success. (Author correspondence: sonyper@stlawu.edu)

Academic performance, Alcohol, Chronotype, Class start times, College, Sleep schedule

INTRODUCTION

Sleep need in puberty remains static at ~9 h for adolescents (Carskadon et al., 1980), whereas the timing of sleep—circadian preference—is delayed (Carskadon et al., 1993, 1997). Delayed circadian preference is thought to be a marker of adolescence (Roenneberg et al., 2004); yet, school start times are set earlier as students progress through public school systems in the United States (Wahlstrom, 2002). As a result, students report getting less sleep than they need from about the age of 13 yrs (Wolfson & Carskadon, 1998). School start times, in part, determine how long middle and high school students sleep; when school starts later, students sleep longer (Carskadon et al., 1995; Owens et al., 2010; Szymczak et al., 1993; Wahlstrom, 2002; Wolfson et al., 2007). Furthermore, when students do not have to be at school, e.g., on weekends, most choose later bedtimes and sleep longer (Wolfson & Carskadon, 1998).

In delaying sleep phase on weekends, students' circadian clock mechanisms are "delayed," meaning that the preference for later bedtimes is exacerbated (Crowley & Carskadon, 2010; for similar findings in adults, see Taylor et al., 2008; Yang et al., 2010). The resulting mismatch between preferred and required schedules is intensified in this weekly pattern. Students who

experience this pattern report increased sleepiness, decreased motivation, more depressed mood states, increased truancy and tardiness, increased dropout rates, and problems with mood regulation (Dahl & Lewin, 2002; Manber et al., 1996; Wolfson & Carskadon, 1998).

Some attempts have been made to address this mismatch of sleep schedules and unmet sleep need. For example, regularizing students' sleep habits and schedules increases sleep quality and length (Manber et al., 1996). In a more direct approach, some school districts have delayed school start times, leading to gains for students across a number of domains (Owens et al., 2010; Wahlstrom, 2002). Fatigue, depression symptoms, and stress reports decrease and truancy and late arrivals likewise decrease, while in some cases average grades increase, albeit slightly (Owens et al., 2010; Wahlstrom, 2002; Wolfson et al., 2007).

Similar to younger students, college students report getting less sleep than they need and indicate that their sleep is often of low quality and irregular (Carney et al., 2006; Gomes et al., 2010; Manber et al., 1996). It is important to know, however, whether *class* start times (in college students) operate in parallel fashion to *school* start times. In a pilot study conducted on university

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Address correspondence to Serge V. Onyper, Department of Psychology, St. Lawrence University, Canton, NY 13617, USA. Tel.: +1 315 229 5121; E-mail: sonyper@stlawu.edu

medical students in Brazil, Lima and colleagues (2002) found that delaying classes by 2.5 h resulted in a delayed sleep onset, a 30-min increase in sleep duration, and better-quality sleep as measured by the Pittsburgh Sleep Quality Index (PSQI). Another study examined morningness-eveningness (ME) preferences in the context of “teaching periods” in which Turkish students enrolled either in courses that occurred during the morning to mid-day period, or courses held during a late afternoon to evening period (Beşoluk et al., 2011). Students with morning preference had superior academic outcomes on final examinations, which were scheduled at 09:30 h for all students; ME scores were significantly related to academic achievement.

Students in this age group have several dynamic systems that are in flux for at least several years after graduation from high school: for example, the nervous system in late adolescence continues to develop into the early 20s (Casey et al., 2008; Galvan et al., 2006), and circadian preference does not reach adult phenotype until about 22 yrs of age (Roenneberg et al., 2004). The question of whether early class start times in this age group might disadvantage learning, behavior, and sleep has not been adequately addressed. In examining the variables that are important in considering how sleep and class schedules might be related in college students, sleep requirements, substance use, daytime behaviors, and cognitive functioning and academics emerge as foci.

Sleep Requirements

Sleep needs in the late adolescent age group of 18–23 yrs are estimated to be about 8.4 h (Wehr, 1991). College students report that they get between 6.1 and 7.4 h sleep/night (Pilcher & Ott, 1998; Pilcher et al., 1996). Singleton and Wolfson (2009) indicated that only 24% of the students in their survey reported adequate sleep, and that delays of the sleep period were common, especially for men.

Alcohol and Substance Use

College students report using more alcohol, caffeine, and tobacco than high school students (White et al., 2005, 2006), and also more illegal substances, including marijuana, depressant substances, and stimulants, and more prescription drugs, including sleep medication. In general, substance use disrupts sleep length and quality, and so increased substance use in college occurs in parallel with decreased sleep quality (Galam-bos et al., 2009; Jacobus et al., 2009; Morin & Wooten, 1996). Most studies have found that excessive, frequent alcohol use almost invariably leads to poor academic achievement (e.g., Singleton & Wolfson, 2009).

Daytime Sleepiness

Irregular, short sleep coupled with a misalignment between circadian preference and daytime schedules combine to worsen sleep and increase sleepiness. Many students regularly report sleepiness (Buboltz

et al., 2006; Wolfson, 2002). Students who get irregular sleep or report less sleep than needed are often tired (Manber et al., 1996) and complain of decreased health (Pilcher & Ott, 1998; Pilcher et al., 1996), especially problems with mental health and mood regulation (Dahl & Lewin, 2002). Thus, the cost of lost sleep is high for students.

Cognitive Functioning and Academic Achievement

Students who sleep less, have more irregular sleep, and report the latest bedtimes typically have lower grade point average (GPA; Kelly et al., 2001; Trockel et al., 2000; Wolfson & Carskadon, 1998, 2003), but the effect sizes can be small. Other decrements appear in sustained attention (Kamdar et al., 2004; Lim & Dinges, 2008), critical thinking (Pilcher & Walters, 1997), problem solving (Campos-Morales et al., 2005; Wagner et al., 2004), and overall cognitive ability (Buboltz et al., 2006). Students with an evening circadian preference find it particularly challenging to perform well in morning schedules (Fernandez-Mendoza et al., 2010; Guthrie et al., 1995). In general, therefore, unmet sleep needs can affect a range of behaviors critical to academic success. In high school, for example, when school schedules started later, truancy rates, absentee rates, and late arrivals decreased and graduation rates improved (Wahlstrom, 2002).

Hypotheses Evaluated in the Current Study

In the present study, we examined first whether class schedules affect sleep and sleep-related behaviors in college to the same extent as they do in middle and high school—that is, whether students with later class start times experience benefits with respect to sleep and daytime functioning. In particular, we expected that students with later class schedules would experience longer and better-quality sleep as well as reduced daytime sleepiness, as compared to students with earlier class schedules. Second, we considered the role of class starting times in academic performance and interactions with other factors, such as sleep habits or substance use. In addition to providing a description of the effects of class start times on sleep, daytime functioning, and academics, we modeled the relationships among these variables using structural equation modeling (SEM).

METHODS

Participants

Two hundred fifty-five participants completed questionnaires on sleep, behavior, mood, health, and substance use, as well as a sleep survey and tests of cognitive function. Participants were undergraduate students at a liberal arts college in the northeastern United States (60% female; the vast majority resided on campus and held no outside jobs) who received credit for participation and provided consent for their study and for

accessing their GPA from the registrar. All participants completed measures between 18:00 and 21:00 h in groups of 12–15, on weekdays in November during the fall semester. The data of two participants were excluded as unusable. The study was approved by the university ethics board and conformed to international ethical standards (Portaluppi et al., 2010).

Materials

Retrospective Sleep Survey

Participants completed a self-report, retrospective sleep survey for the week prior to the testing day. To increase the reliability and validity of the sleep reports, participants were guided through a timeline follow-back procedure similar to that used in other behavioral health literature (LaBrie et al., 2005; Sobell & Sobell, 1992), establishing class schedules, social events, etc., after which participants noted bedtimes, rise times, and first activities after waking.

Questionnaires

Participants completed a measure of circadian preference (the Owl-Lark Scale; Horne & Östberg, 1977) in which they identified preferred schedules for daily activities (higher scores indicate morning preference; Cronbach's $\alpha = .81$). The Pittsburgh Sleep Quality Index (PSQI; Buysse et al., 1989) assessed sleep quality and sleep complaints (reliability of the component scores is .83 in the population and .63 in our sample). Higher scores on PSQI denote more sleep problems. Participants also completed a questionnaire of daytime sleepiness and sleep-wake behavior problems adapted from Carskadon et al. (1991; see also Wolfson & Carskadon, 1998). The questions asked whether in the past 2 wks participants struggled to stay awake or had fallen asleep during 12 different situations (e.g., while studying, watching television, conversing, or traveling), as well as how often in the last 2 wks the participants arrived to class late, felt sleepy during the day, slept in past noon, stayed up all night, had trouble falling asleep, or had problematic sleep (e.g., snored, had nightmares, woke up during the night). The sum of the standardized responses to each question formed the Daytime Sleepiness index (Cronbach's $\alpha = .76$; this composite was created to increase reliability of the measure, an important component of measures used for SEM), with higher scores corresponding to increased sleepiness. Mood was assessed using the Depression, Anxiety, and Stress scale (Lovibond & Lovibond, 1995; Cronbach's $\alpha = .94$; higher values indicate mood complaints).

Participants also completed a brief health questionnaire that asked about incidence (since the start of the semester) of cold and flu, doctor, hospital, or health center visits, and number of classes missed. Students also rated overall health on a 6-point scale and reported whether they abstained from alcohol or were a light, moderate, or heavy drinker. Questions about frequency

of alcohol consumption (from *never* to *daily*, converted into times/yr), number of drinks consumed on a typical weekend night, and binge drinking in the past 2 wks, defined as ≥ 4 drinks within a couple of hours for women (5 for men), were also included. One drink was defined as 12 oz. of beer, 5 oz. of wine, or 1 oz. of liquor. Frequency of use of other substances (caffeine, tobacco, energy drinks, and prescription and nonprescription stimulants) was also included, on a scale from *never* to *daily*. Responses were converted to times/yr. In cases where data on alcohol use frequency, amount, or binge drinking were missing (3% of all responses), mean values for abstainer/light/moderate/heavy drinker groups were imputed, based on alcohol use status (which all participants indicated).

Finally, participants estimated their cumulative GPA, and 80% of the sample provided consent to obtain the current semester's GPA from official records after the end of the semester.

Tests of Cognitive Function

To assess cognitive performance, participants completed a delayed memory recall task (recalling a list of 30 common nouns 10 min after they were studied), Digit-Symbol Coding Task (DSC; Wechsler, 1997), a measure of perceptual-motor speed, and Letter Cancellation Task (LCT; Casagrande et al., 1997), a measure of sustained attention. Participants also rated the overall quality of their memory on a 7-point scale. All cognitive tasks were completed at the beginning of testing. Scores on the three performance measures and subjective memory were standardized and summed to form an index of cognitive performance to create a more reliable measure of cognitive function for inclusion in path analyses.

Path Modeling

Previous research delineated a number of associations among class start times, sleep, daytime functioning, and academic achievement. The current study used SEM to confirm and extend those predictions to relationships between class start times, sleep, alcohol and substance use, and academic performance in a sample of college students. SEM is a confirmatory analytic technique that allows testing of hypothesized causal priority of effects among variables based on underlying theory and existing empirical findings (Kline, 2011). AMOS version 18.0 (Arbuckle, 2009) was used to specify two structural models (i.e., path analyses), one for the relationship between class start times and sleep (in the full sample of participants), and the other for the relationship among class start times, sleep, and academic performance (for participants who released their semester GPA). In each case, modeling began with a specification of an initial model to which additional paths were added to improve model fit (Kline, 2011). Goodness-of-fit was assessed by a chi-square exact-fit test; correlation residuals were inspected to diagnose sources of model

misfit. Models were compared using a chi-square difference test (if nested) or the Bayesian information criterion (BIC). Additionally, the following fit indices were reported (values in parentheses indicate favorable fit; Kline, 2011): Jöreskog-Sörbom goodness-of-fit index (GFI $\geq .95$), Bentler comparative fit index (CFI $\geq .95$), root mean square error of approximation (RMSEA $\leq .05$) with 90% confidence interval (upper bound $< .10$ to reject the poor-fit hypothesis), BIC, and standardized root mean square residual (SRMR $< .08$). The associations among variables are reported as unstandardized (b) and standardized (β) path coefficients that are interpreted in the same manner as coefficients in linear regression (i.e., the change in the criterion variable as a function of one-unit change in the associated predictor, controlling for the other predictors).

RESULTS

Sample Characteristics

The sample consisted of 19% first-year students, 38% sophomores, 21% juniors, and 22% seniors (age: 18–23 yrs). The earliest class start time was 08:00 h; 66% of the sample had at least one class starting between 08:00 and 08:30 h (mean $\pm$ SD: 10:04 h $\pm$ 58 min). On weekdays, participants identified “attending class” as the first scheduled activity of the day in 84% of reported instances; in those cases, classes started >1 h earlier (09:39 h) compared to 16% when class was not the first activity of the day (10:54 h). Although students who had other activities scheduled prior to class had later class start times, they also began their day 37 min earlier than those who did not report any scheduled activities prior to class. These data suggest that even though later class schedules might enable students to seek activities prior to class, relatively few took advantage of this.

Sleep and Chronotype

Responses on the PSQI revealed mean $\pm$ SD bedtimes, rise times, and total sleep times of 00:13 h $\pm$ 55 min, 08:32 h $\pm$ 68 min, and 7.44 h $\pm$ 66 min, respectively, with mean sleep latency of 23 $\pm$ 20 min reported over the

past month. These statistics are consistent with previous reports (e.g., Singleton & Wolfson, 2009). Table 1 presents an overview of the sleep patterns during the week prior to testing.

Correlations between PSQI and sleep survey estimates were high (r s .58 to .78, all p s $< .001$). In addition, 43% of the sample reported weekend bedtime delays of >1 h, and 53% reported weekend oversleep of >1 h based on the retrospective survey estimates; 13.4% reported going at least one night without sleep the previous 2 wks. Men indicated later bedtimes and rise times compared to women on both weekdays and weekends, but there were no sex differences in sleep duration, bedtime delay, or weekend oversleep. The average $\pm$ SD Owl-Lark score in our sample was 49.2 ± 8.7 ; the scores did not differ by sex. There were 41 larks (scores: 59–86) and 49 owls (scores: 16–41), with the rest of the participants falling in the neutral range. Compared to owls, larks reported earlier bedtimes, $t_{88} = 6.72$, $p < .001$ (79 min earlier), and rise times, $t_{88} = 5.74$, $p < .001$ (80 min earlier), but did not differ in PSQI-estimated sleep duration, $t_{88} = 1.07$, $p = .29$. Sleep survey measures yielded comparable results. Interestingly, we also found no evidence supporting preference of larks for morning class schedules: the difference in average class start times between larks and owls was not significant, $t_{88} = 1.20$, $p = .24$, which suggests that although larks may wake up earlier, they do not (or are unable to) enroll in early-morning classes.

Academic Achievement

The self-reported cumulative GPA in the full sample was $3.24 \pm .40$. The current semester's GPA was $3.24 \pm .48$ for the 203 students who released their official records; it correlated highly with their estimated cumulative GPA ($r = .61$, $p < .001$). Men's GPA was significantly lower than women's, both cumulative ($M_M = 3.12 \pm .42$, $M_F = 3.32 \pm .38$, $t_{251} = 4.00$, $p < .001$) and for the semester ($M_M = 3.07 \pm .52$, $M_F = 3.35 \pm .42$, $t_{201} = 4.21$, $p < .001$). A Sex $\times$ Class Year analysis of variance (ANOVA) confirmed the main effect of sex on semester GPA, $F_{1,195} = 10.47$, $p = .001$, $\eta^2 = .05$, but revealed neither a main effect of

TABLE 1. Retrospective survey estimates of sleep during the week prior to testing

Variable	Weekday						Weekend					
	Men		Women		Overall		Men		Women		Overall	
	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD	Mean	SD
Bedtime (h:min)	01:09	61	00:39	60	00:51	62*	01:52	76	01:24	75	01:35	77*
Rise Time (h:min)	08:35	60	08:26	53	08:35	57*	10:31	85	09:59	81	10:12	84*
Sleep Duration (h)	7.83	64	7.89	74	7.87	70	8.29	72	8.17	88	8.20	82
Bedtime Delay (min)	—	—	—	—	—	—	63	48	63	48	63	48
Oversleep (min)	—	—	—	—	—	—	79	66	79	66	79	66

Sleep duration was estimated as a period of time from shutting the eyes with intent to go to sleep until the time the participants awoke and did not close their eyes to go back to sleep; as such, it might overestimate the actual amount of sleep obtained, as time awake during the night was not part of the calculations (i.e., time awake was not subtracted from time in bed). * $p < .05$ for the difference between males and females.

class year, $F_{3,195} = .96$, $p = .41$, $\eta^2 = .02$, nor an interaction, $F_{3,195} = .44$, $p = .73$, $\eta^2 = .01$ (reported partial eta-squared [η^2] represent effect sizes; Cohen [1988] suggested the following cutoff points: .01 for a small effect, .059 for a medium effect, .138 for a large effect).

Mood

Mean scores ($\pm$ SD) for depression, anxiety, and stress (DASS) were 5.20 ($\pm$ 6.00, range 0–35), 5.37 ($\pm$ 5.20, range 0–26), and 9.47 ($\pm$ 7.95, range 0–37), respectively, with 89.3%, 79%, and 86.6% of the sample scoring in the normal-to-mild range, consistent with norms in other student samples (Lovibond & Lovibond, 1995).

Alcohol and Substance Use

Nearly 14% of the sample self-identified as “abstainers,” 33% as “light drinkers,” 47% as “moderate,” and 6% as “heavy” drinkers. Students reported a range of 1–24 drinks on a typical weekend night (mean $\pm$ SD = 5.57 $\pm$ 4.10). In the 2 wks prior to testing, 71% of participants reported ≥ 1 alcohol binges (mean = 2.14 $\pm$ 2.10). Because only 16 participants were “heavy drinkers,” these participants were merged with the “moderate drinker” group. “Abstainers” and “light drinkers” were also combined into a single group for further analyses.

Compared to abstainers/light drinkers, moderate/heavy drinkers were more likely to have evening circadian preference and to report later bedtimes and wake times. They were more likely to sleep less on weekends, use more tobacco and stimulant medication, experience more daytime sleepiness, and have lower scores on cognitive ability measures (all $ps < .05$). Men consumed more alcohol, drank more frequently, and were more likely to binge drink compared to women (all $ps < .05$). Consumption of alcohol, caffeine, tobacco products, energy drinks, and stimulant medication ranged from *never* to *daily*, with women consuming more caffeinated products and men consuming more tobacco products ($ps < .05$). Students who indicated going without sleep for ≥ 1 nights in the 2 wks leading up to testing ($n = 34$) reported significantly more binge drinking and more frequent consumption of energy drinks and stimulants compared to those who did not. Furthermore, their

semester GPA was significantly lower (mean $\pm$ SD = 3.04 $\pm$.44) than that of the students who did not “pull an all-nighter” in the past 2 wks (mean $\pm$ SD = 3.27 $\pm$.47) ($t_{201} = 2.25$, $p = .025$).

Relationship Between Class Start Times and Sleep

We predicted that college students whose schedules included fewer morning classes should experience longer sleep (mainly due to delayed wake times) and reduced sleepiness compared to those with predominantly morning class schedules. The results revealed that students with classes that started later in the day experienced later wake times and longer sleep, but also had later bedtimes. However, the associations between class start times and daytime sleepiness or the ability to remain vigilant and alert (measured by LCT) were not significant. Descriptive statistics and Pearson correlations among key variables of the study are given in the Appendix.

When considering sleep survey responses from the past week, later class start times positively correlated with bedtimes, rise times, and sleep duration on weekdays; on weekends, they correlated with bedtimes only. Moreover, later class start times corresponded to more regular class schedules ($r = .51$, $p < .001$; start time regularity was operationalized as the standard deviation of weekly class start times).

Because recollection by the participant of previous night's sleep may be more reliable than that of sleep a few nights or a week ago, we regressed hours of sleep during the single night leading up to testing on the time of first scheduled class on the day of testing. Controlling for influence of the previous night's sleep, sex, and mood, time of first class was a significant predictor of sleep duration, along with circadian preference and bedtime ($F_{6,231} = 64.95$, $p < .001$; Table 2). This result indicates that 22 min of sleep was gained, on average, for every 1 h that the start of first class was delayed, and this provides preliminary support for our hypothesis that later class start times increase sleep duration.

Next, average weekly class start times were computed for each student, and each participant's schedule was

TABLE 2. Results of multiple regression predicting sleep duration on the night prior to testing

Predictor	<i>b</i>	SE	β	<i>t</i> value
Bedtime	-.912	.051	-.79	17.76*
First class start time	.368	.043	.346	8.55*
Morningness	-.028	.008	-.15	3.46*
Hours of sleep 2 nights ago	-.068	.035	-.078	1.92
Male	.189	.138	.057	1.37
Mood [†]	-.007	.004	-.068	1.66

Morning wake time was not included as a predictor because the criterion variable (sleep duration) is a linear combination of bedtime and wake time. *Regression coefficient significant at $p \leq .05$; $R^2 = .63$; adjusted $R^2 = .62$. All variables were regressed in a single block ($n = 237$). [†]Mood score is the sum total of Depression, Anxiety, and Stress subscale scores of Depression-Anxiety-Stress scale (DASS-42).

TABLE 3. Average weekly class start times, sleep schedules, and alcohol consumption (means $\pm$ SD)

Dependent measure	Class Schedule		
	Early-morning (n = 78)	Mid-morning (n = 127)	Late-morning (n = 48)
Weekdays			
Class Start Time (h:min)	08:56 $\pm$ 20	10:12 $\pm$ 25*	11:29 $\pm$ 23*
Bedtime (h:min)	00:38 $\pm$ 62	00:55 $\pm$ 56	01:01 $\pm$ 74*
Rise Time (h:min)	07:59 $\pm$ 38	08:43 $\pm$ 50*	09:14 $\pm$ 64*
Total Sleep Time (h)	7.40 $\pm$ 1.08	7.97 $\pm$ 1.04*	8.36 $\pm$ 1.36*
Weekends			
Bedtime (h:min)	01:25 $\pm$ 82	01:35 $\pm$ 68	01:51 $\pm$ 86
Rise Time (h:min)	10:09 $\pm$ 88	10:10 $\pm$ 81	10:23 $\pm$ 86
Total Sleep Time (h)	8.38 $\pm$ 1.30	8.14 $\pm$ 1.39	8.15 $\pm$ 1.41
Bedtime Delay (min)	61 $\pm$ 55	66 $\pm$ 44	60 $\pm$ 47
Weekend Oversleep (min)	90 $\pm$ 64	72 $\pm$ 62	80 $\pm$ 77
Alcohol use (Amount)‡	5.08 $\pm$ 3.79	5.49 $\pm$ 3.76	6.56 $\pm$ 5.21*
Alcohol use (Frequency)‡	54.24 $\pm$ 52.60	68.78 $\pm$ 59.36	77.25 $\pm$ 61.69*
Alcohol use (Binge Drinking)‡	1.66 $\pm$ 1.71	2.43 $\pm$ 2.26*	2.17 $\pm$ 2.13

Bedtimes, Rise Times, and Total Sleep Times are given as weighted averages of weekday and weekend times obtained via sleep survey. Bedtime Delay and Weekend Oversleep are deviation scores (i.e., absolute values).

*The group mean is significantly different from the mean of the early-morning group ($p < .05$).

‡Alcohol use was measured as follows: amount = number of drinks on a typical weekend night; frequency = measured on a scale from never to almost daily, with responses converted to times per year; binge drinking = ≥ 4 drinks for women or ≥ 5 drinks for men on a single occasion — that is, within a few hours of each other — over last 2 wks.

then classified as either *early-morning* (with the average start times falling between 08:00 and 09:29 h), *mid-morning* (start times 09:30–10:59 h), or *late-morning* (start times of 11:00 h and later). A series of ANOVAs examined the effect of class schedules on sleep (using retrospective sleep survey data) and other variables. The effect of class schedule on weekday bedtimes failed to reach significance ($F_{2,250} = 2.54$, $p = .08$, $\eta^2 = .02$), although there was a highly significant effect on weekday wake times ($F_{2,250} = 37.00$, $p < .001$, $\eta^2 = .23$) and weekday sleep duration ($F_{2,250} = 11.87$, $p < .001$, $\eta^2 = .09$). On weekday nights, students with late-morning schedules had later wake times compared to the early-morning and the mid-morning groups (who themselves differed significantly), and slept longer than the early-morning group, who had the shortest sleep duration of all three groups (all $ps < .05$; see Table 3). There were no significant differences among groups in weekend bedtimes, rise times, or sleep duration, bedtime delay and oversleep, or circadian preference. Furthermore, course schedules had a significant effect on binge drinking ($F_{2,250} = 3.36$, $p = .036$, $\eta^2 = .03$). Students with early-morning schedules experienced fewer episodes of binge drinking compared to students with mid-morning schedules ($p = .01$), but not students with late-morning schedules ($p = .18$).

An analogous pattern of results emerged when participants were grouped according to the number of days/wk with 08:00 or 08:30 h classes. Those with fewer or no early-morning classes during the week stayed up longer at night, woke up later in the morning, and slept longer

on weekdays (all $ps < .05$). They also consumed alcohol more frequently and had a higher incidence of binge drinking. Thus, class schedule appears to be closely related to sleep during the school week, as well as to alcohol consumption.

Path Analyses: Class Start Times, Sleep, and Academic Performance

Class start times have a significant relationship with sleep habits of college students, and later start times are associated with increased alcohol consumption. A question of particular interest to our study is whether academic performance is related to early morning class schedules, and whether these effects are direct or indirect. For instance, class start times may contribute to altering student sleep schedules, increasing the likelihood of missing classes, or influencing decisions regarding consumption of alcohol and other drugs.

To confirm that class start times affect sleep, which in turn may affect academic performance, we constructed two recursive structural equation models (path analyses). The first model specified relationships between class start times and sleep-related variables, guided by the findings from the adolescent literature (Figure 1). The second model included sleep and several other variables that were thought to mediate the effect of class start times on academic performance (Figure 2). Both models used estimates of bedtimes, wake times, and total sleep time from the PSQI, because its scores reflected sleep-related behaviors over a 1-mo period and because of the well-established reliability of this measure.

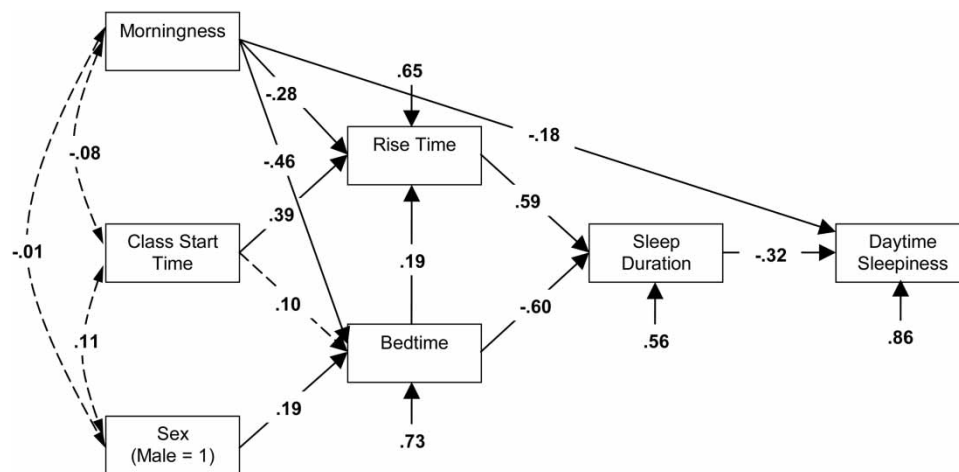


FIGURE 1. Path analysis for the effects of class start times on sleep (Model 1). All direct paths are statistically significant (solid lines) except for the path from Class Start Times to Bedtimes (dashed line; $p = .07$). Covariances among exogenous variables are not statistically significant. Depicted coefficients are standardized; disturbance terms reflect unexplained variance (i.e., $1 - R^2_{SMC}$).

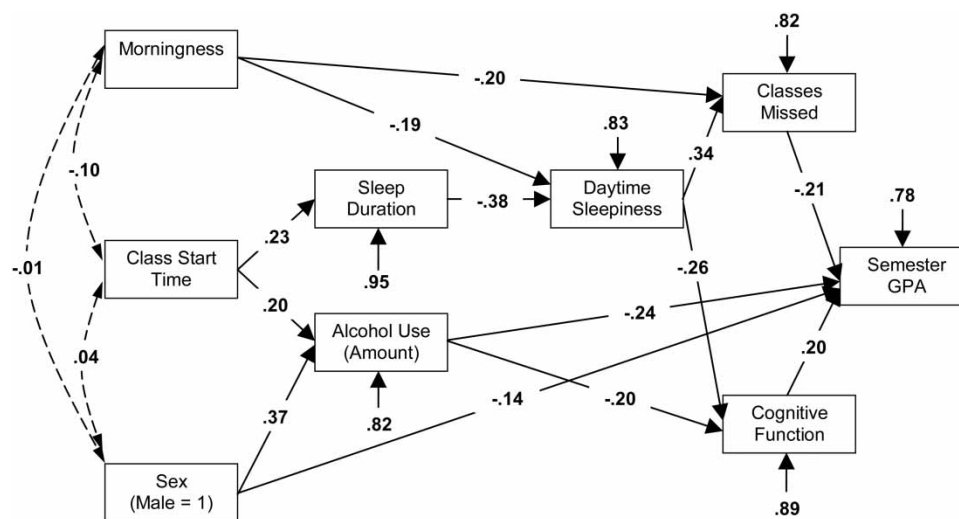


FIGURE 2. Path analysis for the effects of class start times on academic performance (Model 2). All direct effects are statistically significant (solid lines; $p < .01$; $p < .05$ for the path from Sex to Semester GPA). Covariances are not significant (dashed lines). Depicted coefficients are standardized; disturbance terms reflect unexplained variance (i.e., $1 - R^2_{SMC}$).

Model 1: Predicting Sleep-Related Behaviors

Numerous studies have concluded that in middle and high school, delaying school start times often leads to later rise times (and sometimes earlier bedtimes, e.g., Owens et al., 2010; there is also evidence of association between delayed class start times and later bedtimes in university students; Lima et al., 2002), longer sleep, better sleep quality, and less daytime sleepiness (Wahlstrom, 2002; Wolfson & Carskadon, 1998; Wolfson et al., 2007). Thus, we hypothesized that a similar pattern of results would emerge in college. Specifically, we predicted that class start times would directly affect the timing of sleep and indirectly affect sleep duration, by advancing or delaying bedtimes and wake times. We also predicted that longer self-reported sleep would decrease subjective sleepiness during the day. Finally, we included Owl-Lark scores and sex in the model as exogenous variables that we predicted would directly affect the timing of sleep, based on research that

shows morningness predicts earlier bedtimes and rise times (e.g., Guthrie et al., 1995), and that adolescent boys may be more likely than girls to delay bedtime on both weekdays and weekends, in part due to a stronger evening preference (Roenneberg et al., 2004).

The model specified on the basis of these prior predictions (Model 1a, see Table 4) fit the data poorly, and was modified using model-building techniques (Kline, 2011) by adding paths from bedtime to rise time (Model 1b) and from morningness to daytime sleepiness (Model 1c). Both modifications are reasonable; our sample yielded high correlations between bedtime and rise time ($r = .41$, $p < .001$), and between morningness and decreased sleepiness ($r = -.22$, $p < .001$). Thus, those who fall asleep later are also likely to delay their wake time if their schedule permits. Furthermore, according to Medeiros et al. (2001), individuals with evening or moderate evening preference tend to experience poorer

TABLE 4. Fit indices for path analyses of sleep factors (Model 1) and academic performance (Model 2)

Model	Exact fit test			GFI	CFI	RMSEA				SRMR	$\Delta\chi^2$	Δdf	<i>P</i>
	χ^2	<i>P</i>	<i>df</i>			estimate	90% CI	<i>p</i> _{close fit}	BIC				
Model 1 (Class start times & Sleep, <i>N</i> = 253)													
Model 1a (initial model)	30.76	<0.01	10	.97	.94	.09	.06 to .13	.03	130.4	.07			
Model 1b (added: bedtime → rise time)	20.58	.02	9	.98	.97	.07	.03 to .11	.17	125.7	.05	10.18	1	<0.02
Model 1c(added: morningness → daytime sleepiness)	11.53	.17	8	.99	.99	.04	.00 to .09	.55	122.2	.03	9.05	1	< 0.03
Model 2 (Class start times, Sleep, & GPA; <i>N</i> = 203)													
Model 2a (Model c: substituting sleep midpoint for bed/risetimes)	18.58	.01	6	.97	.91	.1	.05 to .16	.05	98.3	.06			
Modle 2b (added: class start times → sleep duration)	6.09	.3	5	.99	.99	.03	.00 to .11	.56	91.1	.04	12.49	1	<.001
Model 2c (trimmed: sleep midpoint → sleep duration)	7.53	.27	6	.99	.99	.01	.00 to .10	.56	87.2	.04	1.45	1	.23
Model 2d (Model 2c + alcohol use, classes missed, ability, GPA)	64.28	<.001	27	.94	.87	.08	.06 to .11	.02	213.1	.08			
Model 2e (added: sleepiness → ability, morningness → missed classes, class start times → alcohol use)	31.67	.14	24	.97	.97	.04	.00 to .07	.65	196.4	.05	32.61	3	<.001
Model 2f (trimmed: alcohol use → sleep duration, alcohol use → sleep midpoint, alcohol use → classes missed, sleep midpoint)	30.36	.09	21	.97	.95	.05	.00 to .08	.52	157.9	.06			
Model 2g (added: gender → GPA)	25.99	.17	20	.97	.97	.04	.00 to .08	.65	158.8	.05	4.37	1	.04

All models are presented in the order in which they were run. Addition/trimming of paths or variables, as well as the change in the improvement of fit $\Delta\chi^2$, are in reference to the preceding model. Boldface = accepted final model shown in Figure 1 (Model 1, Class Start Times and Sleep) and Figure 2 (Class Start Times, Sleep, and Academic Performance).

GFI = Jöreskog - Sörbom goodness-of-fit index; CFI = Bentler comparative fit index; RMSEA = Steiger-Lind root mean square error of approximation; *p*_{close fit} = *p* value for the close fit hypothesis of RMSEA; BIC = Bayesian information criterion; SRMR = standardized root mean square residual.

TABLE 5. Maximum likelihood estimates for final models of sleep factors (Model 1) and academic performance (Model 2)

Parameter	Model 1 (Sleep factors)		
	Unstandardized	SE	Standardized
Direct effects			
Class start time → Bedtime	.09	.05	.10
Class start time → Rise Time	.46	.06	.39
Morningness → Bedtime	−.05	.01	−.46
Morningness → Rise Time	−.04	.01	−.28
Sex → Bedtime	.35	.10	.19
Bedtime → Rise Time	.23	.07	.19
Rise Time → Sleep Duration	.57	.05	.59
Bedtime → Sleep Duration	−.72	.06	−.60
Sleep Duration → Sleepiness	−3.27	.60	−.32
Morningness → Sleepiness	−.23	.08	−.18
Disturbance variances			
Bedtime	.61	.05	.73
Rise Time	.84	.08	.65
Sleep Duration	.68	.06	.56
Sleepiness	108.51	9.67	.86
Parameter	Model 2 (Academic performance factors)		
	Unstandardized	SE	Standardized
Direct effects			
Class start time → Sleep Duration	.25	.08	.23
Class start time → Alcohol Use	.84	.27	.20
Morningness → Sleepiness	−.24	.08	−.19
Morningness → Classes Missed	−.02	.01	−.20
Sex → Alcohol Use	3.09	.54	.37
Sex → GPA	−.14*	.07	−.14
Sleep Duration → Sleepiness	−3.98	.68	−.38
Sleepiness → Classes Missed	.02	.00	.34
Sleepiness → Cognitive Function	−.06	.02	−.26
Alcohol Use → Cognitive Function	−.13	.04	−.20
Alcohol Use → GPA	−.03	.01	−.24
Classes Missed → GPA	−.14	.04	−.21
Cognition → GPA	.04	.01	.20
Disturbance variances			
Classes Missed	.38	.04	.82
Sleep Duration	1.07	.11	.95
Sleepiness	103.62	10.31	.83
Alcohol Use	13.87	1.38	.82
Cognition	6.07	.60	.89
GPA	.17	.02	.78

Disturbance estimates reflect proportions of unexplained variance. Parameters in **bold** are statistically significant at $p < .01$. * $p < .05$.

sleep and increased daytime dysfunction, especially if they have early morning work or study schedules. Model 1c resulted in a good overall fit ($\chi^2_8 = 11.53$, $p = .17$, GFI = .99, RMSEA = .04 [90% CI = .00 to .09, $p = .55$], SRMR = .03 [all correlation residuals < .10]). All path coefficients were significant ($ps < .002$), except for the path from class start times to bedtimes ($b = .10$, $\beta = .10$, $p = .071$). That path was retained, however, because it is in agreement (i.e., in establishing a weak but direct relationship) with the only other estimate of association between class start times and bedtimes for college populations (Lima et al., 2002). Furthermore, in studies of middle and high school students this effect is not consistently found, e.g., Owens et al. (2010) found

that later school start times were associated with somewhat *earlier* bedtimes; thus, its retention may encourage replication in future research.

The final accepted model (Model 1c) of the effects of class start time on sleep factors is shown in Figure 1. Unstandardized estimates of direct effects and disturbance variances are presented in Table 5. Total, i.e., sum of direct and indirect, effects on each endogenous variable are shown in Table 6. The findings from this path analysis provide support for the hypothesis that class start times exert a powerful influence on sleep in college-aged adults. Similar to middle and high school student populations, later class start times in our sample were associated with later rise times, later

TABLE 6. Standardized total effects on endogenous variables in Model 1 (Sleep factors) and Model 2 (Academic performance factors)

Model 1 (Sleep factors)						
Endogenous variable	Causal variable					
	Sex	Class Start Time	Morningness	Bedtime	Rise Time	Sleep Duration
Bedtime	.19	.10	-.46	—	—	—
Rise Time	.04	.41	-.36	.19	—	—
Sleep Duration	-.09	.19	.06	-.49	.59	—
Daytime Sleepiness	.03	-.06	-.20	.16	-.19	-.32

Model 2 (Academic performance factors)								
Endogenous variable	Causal variable							
	Sex	Class Start Time	Morningness	Sleep Duration	Alcohol use	Daytime Sleepiness	Cognitive Function	Classes Missed
Sleep Duration	—	.23	—	—	—	—	—	—
Alcohol Use	.37	.20	—	—	—	—	—	—
Daytime Sleepiness	—	-.09	-.19	-.38	—	—	—	—
Cognitive Function	-.07	-.02	.05	.10	-.20	-.26	—	—
Classes Missed	—	-.03	-.26	-.13	—	.34	—	—
GPA	-.24	-.04*	.06	.05	-.28	-.12	.20	-.21

Bootstrapping ($n = 1000$) was used to estimate standard errors for standardized total effects; **bold** = bias-corrected two-tailed significance $< .01$; $*p = .03$. We chose to report only standardized effect coefficients to ease the interpretability of results. Note that the statistical significance of unstandardized effects paralleled that of the standardized effects depicted here.

(although not by a statistically significant amount) bedtimes, and longer sleep overall. Furthermore, those participants who reported later class start times experienced less daytime sleepiness as a result of increased sleep duration. Participants with earlier bedtimes slept more, as did those with later rise times.¹ Additionally, men went to bed later than women, and morningness was associated with earlier bedtimes, earlier rise times, and less daytime sleepiness overall.

Model 2: Predicting Academic Performance

Next, we modeled associations between class start time, sleep, and academic performance, focusing on a subset of participants ($n = 203$) whose semester GPA was obtained from official records. As a starting point, we modified the model shown in Figure 1 by combining the rise time and bedtime variables into a measure of sleep midpoint to anticipate an increase in model complexity associated with adding additional variables. The fit of the resulting model, however, was unsatisfactory (Model 2a, see Table 4), and thus a path from average class start times to sleep duration was added (Model 2b), suggesting that class start times may have a direct effect on sleep amount, in addition to an indirect effect through sleep midpoint. The resulting model passed the exact-fit test, although the coefficient for the path from sleep midpoint to sleep duration was no

longer significant ($b = .11$, $p = .23$). Consequently, that path was eliminated ($\chi^2_1 = 1.45$, $p = .23$).

The fit of the resulting, more parsimonious, model (Model 2c) was acceptable ($\chi^2_6 = 7.53$, $p = .27$, GFI = .99), CFI = .99, RMSEA = .04 [90% CI = .00 to .10, $p = .56$], SRMR = .04). The difference in BIC between Model 2a and Model 2c (11 points; the models are not nested) favors Model 2c (Raftery [1993] suggests that a difference of >10 points indicates strong preference for one model over another). Thus, Model 2c formed the basis for adding additional variables, including semester GPA.

Four additional endogenous variables were added to the base model: alcohol consumed on a typical weekend night, classes missed, cognitive ability, and semester GPA. Previous research (e.g., Singleton & Wolfson, 2009) suggests that men consume more alcohol, and that alcohol amount consumed by college students on a typical weekend night is correlated with later sleep midpoints and reduced sleep duration. Increased alcohol use is also linked to lower academic performance (Presley & Pimentel, 2006; Singleton, 2007; Wechsler et al., 2000; Wolaver, 2002). This research provides the basis for adding paths from sex to alcohol consumption, and from alcohol consumption to sleep midpoint, sleep duration, and GPA.

We also know from studies of middle and high school students that delayed school start times are associated

¹There is evidence of a suppression effect between bedtime and sleep duration. While the direct effect of bedtime on sleep duration is negative, i.e., later bedtimes tend to result in less total sleep, the indirect effect—via rise times—is positive. That is, later bedtimes may also lead to later rise times, which are associated with more sleep overall. The former outcome is more likely on weekdays or days that have early morning activities, the latter on weekends or days when no early morning activities are scheduled.

with improved attendance, mediated by improved sleep (Owens et al., 2010; Wahlstrom, 2002; Wolfson et al., 2007). Extending these results to our study, we would predict an indirect effect of class start times on attendance, by increasing the amount of sleep and correspondingly decreasing sleepiness and daytime dysfunction; thus, a path from sleepiness to classes missed can be added.

We can also predict that both class attendance and cognitive ability would have a direct effect on GPA, and that alcohol use would affect GPA indirectly, by worsening memory and reducing learning (Singleton & Wolfson, 2009; Ziegler et al., 2005) and by contributing to absences (Wechsler et al., 2000). Therefore, we added paths from classes missed to GPA, as well as from alcohol use to cognitive ability to GPA.

The fit of the resulting model was unsatisfactory (Model 2d; see Table 4), with many correlation residuals $>.10$. Examination of residuals led to adding additional pathways, from daytime sleepiness to cognitive function (see Campos-Morales et al., 2004; Kamdar et al., 2004; see also Buboltz et al., 2006; Pilcher & Walters, 1997), from chronotype to missed classes (see Guthrie et al., 1995; Medeiros et al., 2001), and from class start times to alcohol consumption (Model 2e). This model resulted in a significant improvement in fit over Model 2d, but could be made more parsimonious, by removing paths from alcohol consumption to sleep duration ($b = -.002$, $\beta = -.01$, $p = .91$), sleep midpoint ($b = -.001$, $\beta = -.01$, $p = .92$), and missed classes ($b = .01$, $\beta = .06$, $p = .32$), as well as omitting the endogenous variable of sleep midpoint. The resulting model (Model 2f) provided a reasonable fit to the data, but also underestimated the observed correlation between sex and GPA, suggesting a direct sex effect. Adding that final path resulted in a significant improvement in the goodness-of-fit and an excellent model fit overall (Model 2g; with all correlation residuals $\leq .11$). Thus, Model 2g was accepted as the final model relating class start times, sleep, and academic performance (Figure 2).

Path Models and Study Hypotheses

The examination of path coefficients (direct effects) for Model 1c (Figure 1) and Model 2g (Figure 2) and the total effects on endogenous variables reported in Table 6 allows us to check modeling outcomes against our predictions.

Our first hypothesis postulated a link between class start times and sleep. Model 1c revealed that an increase in sleep duration associated with later class start times is due mainly to delayed rise times. Model 1c and Model 2g both show that class start times also had an indirect effect on daytime sleepiness, i.e., later classes resulted in more sleep, which, in turn, was associated with reduced sleepiness.

Our second hypothesis predicted that, because class start times would affect sleep, they would also influence academic achievement. Neither of the two sleep variables included in Model 2g had a direct effect on GPA, but both affected it indirectly and in a way consistent with previous

research and our expectations. Specifically, increased sleep duration was directly associated with reduced daytime sleepiness ($\beta = -.38$) and indirectly associated with improved cognitive function ($\beta = .10$), decreased number of missed classes ($\beta = -.13$), and higher GPA ($\beta = .05$). Attendance and cognitive performance, in turn, were potent, direct predictors of GPA.

The overall effect of class start times on GPA was small and entirely indirect; an increase of 1 SD in average class start time—of ~ 1 h—corresponded to a .044 SD decrease in semester GPA—of $\sim .021$ points. The strongest single predictor of GPA was the amount of alcohol consumed on a typical weekend night ($\beta = -.28$); those with greater alcohol consumption had lower semester GPAs. Furthermore, male sex had a direct negative effect on GPA ($\beta = -.24$), as well as affected GPA indirectly via increased alcohol use. Thus, alcohol use appears to be one of the most important variables contributing to GPA. It is also the factor that differentiates the results of this study from those reported for middle school and high school students.

DISCUSSION

The current research investigated the relationship among class start times, sleep and sleep-related daytime functioning, and academic achievement in college students. College is often seen as a period of transition from adolescence to adulthood (Zimmermann, 2011), and previous research has not adequately addressed whether early class schedules in college students would have effects similar to early school start times in middle and high school students. Our results indicate that student sleep schedules were closely linked to their academic schedules. Having classes that started later in the day was associated with delayed wake times (and slightly later bedtimes), but also longer sleep and reduced daytime sleepiness. Specifically, delaying classes by 1 h resulted in a gain of 22 min of sleep on weekday nights (see Table 2) and 15 min of sleep overall (i.e., the direct effects of class start times for Model 2g, Table 5). Sleep gains were mainly due to a delay in wake times stemming from later class starting times. Similar to middle and high school contexts, delaying start times increases the amount of nighttime sleep, reduces daytime fatigue, and improves attendance (Carskadon et al., 1998; Owens et al., 2010; Wahlstrom, 2002; Wolfson et al., 2007). For both populations, it appears that fatigue may drive part of the decision of whether to attend class: as daytime sleepiness mounts, class attendance decreases. Thus, although students are not missing entire days—they are not “truant”—they are more likely to miss at least some classes, which negatively affects learning and leads to poorer GPA.

Both sleep duration and daytime sleepiness were indirectly related to academic performance; students reporting longer sleep and reduced daytime sleepiness also tended to earn higher grades. The effect of

sleepiness on GPA, however, was almost 3 times as large as the effect of sleep duration (Table 6). Sleepiness affected cognitive function (increased sleepiness corresponded to lower scores on tests of cognitive ability) and attendance (daytime sleepiness was associated with a higher propensity for missing classes, which related inversely to GPA). Daytime sleepiness is clearly bad for academic performance (e.g., Link & Ancoli-Israel, 1995; Rodrigues et al., 2002; Singleton & Wolfson, 2009), but the association between sleep duration and GPA is less reliable. Some studies, both on high school and college students, show evidence for it (e.g., Kelly et al., 2001; Wolfson & Carskadon, 1998), and some argue against it (Eliasson et al., 2002; Gray & Watson, 2002; Howell et al., 2004). These inconsistencies may be due to the small size of the effect. In our findings, the Pearson correlation between sleep duration and GPA was not significant, and the indirect effect of sleep duration on GPA was trivial, with an increase of 1 h in sleep duration corresponding to only a .02 point increase of semester GPA (Table 6).

In short, class schedules are only weakly associated with semester GPA when the influence of other variables is controlled; we found that for every 1 h that classes started later, GPA declined by .022 points. For instance, the GPA of students with predominantly early morning schedules (the “early morning” group, see Table 3) would be almost .06 points higher than the GPA of those with predominantly late schedules (the “late-morning” group). Thus, although later classes enable students to sleep longer, which improves daytime functioning and ultimately has a positive effect on semester GPA, the overall effect of later class start times on academic performance, albeit small, is *negative*. The negative relationship between class start times and GPA contrasts with studies that have found either no statistically significant relationships between school start times and GPA (Owens et al., 2010; Wahlstrom, 2002; Wolfson & Carskadon, 1998), or report increase in academic achievement with later school start times (Wolfson et al., 2007). Importantly, this implies that factors other than sleep or daytime functioning may act as powerful mediators of the relationship between the start time of classes and academic achievement. Our findings suggest that one such factor is alcohol consumption.

Alcohol and substance abuse are prevalent in college, much more so than in secondary school (White et al., 2006). Furthermore, consuming alcohol in excessive amounts has been shown to affect both the quality and duration of sleep (Galambos et al., 2009; Jacobus et al., 2009) and the quality of academic performance (Singleton & Wolfson, 2009). We found that alcohol use was associated both with delayed sleep schedules and lower GPA. Importantly, students with later class schedules were more likely to drink heavily. The number of drinks consumed on a typical weekend night was the strongest predictor of academic performance (Table 6); an increase

of 1 SD in alcohol use (≈ 4 drinks) was associated with a .28 SD decline in semester GPA ($\approx .13$ points). In our second model, average class start time, together with sex, explained 18% of variability in alcohol consumption. These results imply that class schedules may partly govern student substance use.

We should acknowledge that our study measured weekend alcohol consumption only, whereas it is possible that students may also drink on weekday nights. Students may choose to consume alcohol (and stay up later) when their schedule allows, in particular when there is no need to get up early in the morning. Nonetheless, all three measures of alcohol use employed in the study (frequency, amount consumed on weekends, and severity of binge drinking) correlated highly, which suggests that weekend alcohol intake is a reliable predictor of overall alcohol consumption. Notably, weekend alcohol use predicted delayed bedtimes and rise times on both weekends and weekdays, yet correlated with sleep duration on weekends only: students who drink more heavily on weekends may compensate for reduced sleep and perhaps study time lost while drinking by staying up later on weekdays. Their sleep duration on weekday nights, however, is not affected by the amount of alcohol they consume on weekends. Along with delaying bedtimes, students may also delay their wake times—as long as their class schedules allow. Furthermore, at least some of the negative effects of the student's sex on academic performance can be attributed to men's increased alcohol intake.

Another factor that might explain the positive effects of early class start times on GPA is self-selection based on circadian preference. Our results show that circadian preference moderates bedtimes, rise times, and daytime sleepiness. Compared to students with an evening preference, morning types go to bed earlier, wake up earlier, and experience fewer sleep-related problems during the day, including missing class. Research also suggests that students who are morning types often obtain higher grades, possibly because of their superior performance in early-morning classes (Guthrie et al., 1995; Taylor et al., 2011). Thus, it is possible that unlike high school students, those in college have increased autonomy to select class schedules that are consistent with their chronotype. In our sample, however, morning preference correlated only weakly with average class start times ($r = -.08$, *ns*), and there was no evidence that larks preferentially enrolled in earlier classes. This indicates that despite the possibility that increased autonomy, together with different circadian preference, might lead to a divergent class schedule between owls and larks, this was not the case in our study.

On the other hand, chronotype preference correlated reliably with the question regarding how ideal the student's current semester schedule was ($r = .19$, $p < .01$). This suggests that selection of class schedules may have little to do with personal preferences in our sample; yet, if one happened to have a morning circadian preference, a schedule that almost always contained morning classes

was more likely to be well tolerated. This is not surprising, considering that the data were collected in a small (2400 students) liberal arts college with a very limited selection of courses that are offered as multiple sections at different times of day. Thus, students with a predominantly evening preference, if given a choice, would not enroll in early morning classes; yet, if they must enroll, as our data suggest they do, they would still favor their biological tendency to stay up later at night, suffering from increased daytime dysfunction as a result. When this pattern of behavior is combined with increased likelihood of alcohol consumption, enabled by favorable course schedules, academic outcomes may suffer.

These data paint a complex picture of the relationship between academic success, student schedules, and late adolescent physiology. Class schedules, chronotype, habits of daily living—including substance use and abuse—and cognitive ability all mutually influence each other, against a background of the student's sex and health status. Therefore, for students struggling with achievement, assessment of the demands of their schedules and their patterns of substance use are both important.

Although student GPA was collected at the end of the academic semester and can be presumed to follow from the behaviors (e.g., sleep schedules, class start times) that occurred during the semester, the design of the present study was correlational, and most of the variables were collected using self-report. Thus, caution should be used in drawing causal inferences from our findings. Nonetheless, our findings are consistent with those reported by a variety of other sources that used both self-report and objective measures, e.g., in regard to sleep schedules and duration in this population, and internal consistency coefficients for most of our measures are high. Our sample comes from a relatively homogeneous population of students studying at a private, residential liberal arts university in the northeastern United States, and only replication in different populations will be able to demonstrate that our conclusions are generalizable across a larger pool of young adults, both in college and not.

Finally, we should emphasize that the decision about which variables to include in the model was guided, in part, by theory and previous research, but also by the variable's ability to increase the model's predictive power of GPA. To preserve parsimony, variables that did not contribute to additional variance in GPA beyond those already included in the model, e.g., mood, health problems, or use of substances other than alcohol, were omitted from the final path models. It is important to acknowledge, however, that several of them have an important relationship with many other variables considered in the current study, for instance, mood correlated highly with sleep duration, PSQI scores, sleepiness, health problems, cognitive performance, and number of missed classes. Thus, they should be examined more fully in future studies.

CONCLUSION

Our study investigated the relationship between college class start times, sleep and sleep-related daytime dysfunction, and academic performance. Our results corroborate earlier findings from middle and high school students, i.e., college students with later class schedules sleep longer, feel less daytime sleepiness, and miss fewer classes. Furthermore, chronotype moderates sleep schedules and daytime functioning; those with a later circadian preference experience diminished daytime function, including greater sleepiness and increased likelihood of missing class. We also found that academic performance relates only slightly to class start times, and not in a direction reported in studies of adolescents; there was a negative association between class start times and GPA. It may be that students in college are better able to cope with their demanding schedules. For example, they may experience problems stemming from sleep deficits due to earlier class start times, but these problems may be offset by more effective countermeasures or by better-informed decisions regarding habits of daily living (e.g., Zimmermann, 2011). Balancing these influences, however, is the increased exposure to and use of substances, both of which are more pronounced in the older age group. In fact, our results suggest that excessive alcohol consumption constitutes the main negative influence on academic performance in college students. Although class schedules do affect sleep and academic performance, their effects appear to be secondary to those of alcohol use and abuse.

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APPENDIX

Variable	N = 253			Pearson Correlations																
	Mean	SD	Range	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.
1. Average Class Start Time (hr:min)	10:04	58 min	8:30–12:16	1	.512**	.131	.436**	.353**	.231**	.171*	.515**	.298**	.095	-.009	-.141*	-.050	-.011	.010	-.005	-.101
2. Class Start Time Regularity [¶]	1.20	.85	0.00–3.89	.512**	1	.025	.175*	.126	.147*	.006	.196**	.193**	.011	.039	-.015	.026	-.089	-.047	-.003	-.028
3. Bedtimes (PSQI)	00:13	55 min	22:00–4:30	.155*	.037	1	.407**	.808**	-.375**	.777**	.361**	-.493**	.526**	.293**	-.102	-.138*	.183**	.139*	.265**	-.462**
4. Rise Times (PSQI)	08:32	68 min	5:00–12:30	.445**	.162**	.378**	1	.867**	.323**	.359**	.628**	.198**	.274**	.227**	.065	-.079	-.016	.110	.089	-.394**
5. Sleep Midpoint (PSQI)	04:22	52 min	1:45–8:00	.379**	.128*	.787**	.869**	1	.004	.655**	.602**	-.141*	.463**	.306**	-.014	-.127	.089	.147*	.202**	-.506**
6. Sleep Duration (PSQI)	7.44	1.10	4.00–11.00	.229**	.126*	-.376**	.366**	.043	1	-.341**	.246**	.616**	-.216**	-.013	.128	-.060	-.215**	-.494**	-.261**	.071
7. Weekday Bedtime	00:51	62 min	21:48–5:06	.164**	.032	.778**	.334**	.638**	-.340**	1	.399**	-.574**	.552**	.374**	-.206**	-.156*	.191**	.219**	.390**	-.390**
8. Weekday Rise Time	08:35	56 min	5:30–12:01	.535**	.217**	.385**	.641**	.633**	.254**	.400**	1	.425**	.384**	.375**	.039	-.028	-.068	.005	.155*	-.424**
9. Weekday Sleep Duration	7.87	1.17	3.00–10.97	.315**	.180**	-.478**	.244**	-.093	.640**	-.572**	.415**	1	-.278**	-.024	.089	.036	-.296**	-.246**	-.273**	.025
10. Weekend Bedtime	01:35	76 min	21:30–6:15	.144*	.036	.542**	.324**	.506**	-.186**	.559**	.433**	-.246**	1	.527**	-.189**	.419**	.051	.127	.210**	-.360**
11. Weekend Rise Time	10:12	84 min	5:15–15:00	.039	.045	.305**	.252**	.331**	.014	.373**	.439**	.024	.511**	1	.508**	.118	.164*	.077	.055	-.403**
12. Weekend Sleep Duration	8.22	1.37	4.00–12.75	-.096	-.027	-.104	.025	-.039	.113	-.224**	.066	.127*	-.210**	.511**	1	-.086	.207**	.008	-.142*	-.125
13. Bedtime Delay	1.06	.80	0.00–4.45	.032	.052	-.122	.017	-.054	.010	-.167**	.055	.116	.444**	.136*	-.073	1	-.072	.072	.195**	-.002
14. Oversleep	1.32	1.10	0.00–7.09	-.034	-.064	.183**	-.001	.097	-.177**	.181**	-.023	-.262**	.082	.172**	.155*	-.036	1	.281**	.095	-.051
15. PSQI Global Score	6.26	2.92	1–18	.026	-.013	.154*	.125*	.166**	-.483**	.225**	.022	-.242**	.156*	.079	-.024	.075	.269**	1	.132	-.181**
16. All-nighters (1=yes, past 2 wks) [*]	-	-	-	.011	.006	.245**	.095	.194**	-.209**	.354**	.152*	-.242**	.233**	.041	-.186**	.213**	.087	.104	1	-.106
17. Morningness (Owl-Lark score)	49.17	8.74	20–73	-.080	.005	-.470**	-.396**	-.515**	.066	-.413**	-.422**	.048	-.399**	-.414**	-.091	-.010	-.079	-.194**	-.071	1
18. Daytime Sleepiness [§]	.00	11.26	-23.2–40.1	-.037	.010	.177**	.029	.114	-.331**	.249**	.030	-.234**	.180**	.099	-.064	.129*	.198**	.465**	.244**	-.199**
19. Cognition Index [§]	.00	2.55	-5.8–6.8	.000	.102	-.078	-.073	-.090	.080	-.074	-.090	.007	-.099	-.077	.003	-.073	-.090	-.153*	-.077	.076
20. Classes Missed [†]	1.64	.73	1.00–4.58	.039	-.052	.195**	.134*	.193**	-.198**	.229**	.192**	-.115	.238**	.241**	.042	.152*	.170**	.306**	.167**	-.228**
21. Health Problems	4.62	3.17	1–23	.015	-.110	.068	.110	.110	-.077	.078	.121	.002	.012	.044	.020	.053	.098	.324**	.023	-.148*
22. Mood (DASS score)	20.04	16.54	0.00–82.00	-.083	-.100	-.045	-.057	-.062	-.176**	-.027	-.117	-.112	-.032	-.019	.034	.052	.236**	.457**	.057	.015
23. Alcohol Use (Amount)	5.57	4.10	0–24	.181**	.086	.161*	.154*	.189**	.067	.204**	.201**	.030	.254**	.122	-.155*	.064	-.092	-.001	.094	-.113
24. Alcohol Use (Severity)	2.14	2.10	0–12	.173**	.093	.175**	.186**	.218**	.047	.283**	.249**	.010	.343**	.128*	-.260**	.143*	-.064	.004	.231**	-.206**

Continued

APPENDIX Continued

Variable	N = 253			Pearson Correlations																
	Mean	SD	Range	1.	2.	3.	4.	5.	6.	7.	8.	9.	10.	11.	12.	13.	14.	15.	16.	17.
25. Alcohol Use (Frequency)	65.91	58.21	0–365	.208**	.100	.173**	.200**	.226**	.080	.247**	.284**	.037	.293**	.142*	-.141*	.096	-.113	-.010	.115	-.147*
26. Caffeine Use (Frequency)	145.04	145.91	0–365	-.058	-.133*	.087	-.146*	-.051	-.198**	.110	-.101	-.226**	.076	.028	-.037	-.027	.125*	.172**	.001	-.066
27. Tobacco Use (Frequency)	26.73	80.01	0–365	.131*	.066	.157*	.070	.130*	-.012	.138*	.189**	-.028	.199**	.124*	-.003	.059	-.014	.028	.086	-.082
28. Energy Drink Use (Frequency)	18.13	54.42	0–365	.049	-.030	.171**	-.072	.043	-.248**	.121	.068	-.155*	.102	.005	-.089	-.028	.088	.186**	.149*	-.117
29. Stimulant Use (Frequency)	17.74	70.77	0–365	.103	.050	.022	.029	.031	-.055	.011	.024	.004	-.042	-.060	-.064	.152*	.010	.178**	.169**	.053
30. Sex (1=male)*	-	-	-	.101	.056	.187**	.167**	.211**	.022	.202**	.185**	-.025	.163**	.168**	.043	.057	-.084	.015	.159*	-.046
31. Cumulative GPA estimate	3.24	.40	2.00–4.00	-.129*	-.101	-.098	-.129*	-.138*	.032	-.130*	-.214**	-.080	-.122	-.166**	.021	-.016	.093	-.052	-.058	.123
32. Semester GPA (N=203)	3.24	.48	1.13–4.00																	

Variable	N = 253			Pearson Correlations																	N=203	
	Mean	SD	Range	18.	19.	20.	21.	22.	23.	24.	25.	26.	27.	28.	29.	30.	31.	32.	Mean	SD		
1. Average Class Start Time (hr:min)	10:04	58 min	8:30–12:16	-.001	.031	.046	.024	-.056	.212**	.199**	.225**	-.133	.140*	.046	.122	.029	-.132	-.035	10:02	58 min		
2. Class Start Time Regularity [¶]	1.20	.85	0.00–3.89	-.013	.106	-.055	-.114	-.129	.141*	.163*	.156*	-.166*	.065	-.027	.063	.030	-.136	.015	1.20	.84		
3. Bedtimes (PSQI)	00:13	55 min	22:00–4:30	.213**	-.072	.210**	.071	-.034	.207**	.181**	.175*	.064	.149*	.133	.045	.172*	-.129	-.133	00:11	57 min		
4. Rise Times (PSQI)	08:32	68 min	5:00–12:30	.049	-.064	.166*	.118	-.085	.114	.136	.114	-.221**	.041	-.102	.034	.161*	-.118	-.127	08:29	67 min		
5. Sleep Midpoint (PSQI)	04:22	52 min	1:45–8:00	.148*	-.081	.221**	.115	-.074	.186**	.186**	.168*	-.107	.107	.007	.047	.198**	-.146*	-.155*	04:20	52 min		
6. Sleep Duration (PSQI)	7.44	1.10	4.00–11.00	-.388**	.064	-.198**	-.090	-.204**	.041	.022	.044	-.260**	.010	-.226**	-.065	.063	.083	.119	7.39	1.06		
7. Weekday Bedtime	00:51	62 min	21:48–5:06	.284**	-.098	.242**	.099	-.023	.228**	.280**	.243**	.086	.106	.079	.046	.187**	-.149*	-.099	00:49	62 min		
8. Weekday Rise Time	08:35	56 min	5:30–12:01	.052	-.068	.165*	.143*	-.101	.210**	.263**	.267**	-.174*	.206**	.022	.033	.140*	-.200**	-.104	08:31	55 min		
9. Weekday Sleep Duration	7.87	1.17	3.00–10.97	-.258**	.027	-.129	.011	-.104	-.005	.022	.018	-.258**	-.005	-.119	-.013	-.054	-.046	.037	7.83	1.15		
10. Weekend Bedtime	01:35	76 min	21:30–6:15	.188**	-.101	.232**	-.018	-.037	.253**	.303**	.281**	.069	.206**	.064	-.074	.131	-.140*	-.173*	01:32	76 min		
11. Weekend Rise Time	10:12	84 min	5:15–15:00	.079	-.082	.235**	.066	-.006	.148*	.157*	.147*	.026	.144*	-.041	-.059	.157*	-.210**	-.128	10:11	84 min		
12. Weekend Sleep Duration	8.22	1.37	4.00–12.75	-.061	.014	.059	.047	.087	-.128	-.223**	-.145*	-.023	.021	-.105	-.072	.047	-.006	-.024	8.27	1.36		
13. Bedtime Delay	1.06	.80	0.00–4.45	.122	-.054	.127	.029	.034	.020	.082	.050	-.050	.067	-.042	.135	.036	-.047	-.122	1.05	.80		

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14. Oversleep	1.32	1.10	0.00–7.09	.169*	–.103	.197**	.110	.222**	–.026	–.010	–.030	.155*	.012	.040	.049	–.037	.086	.015	1.36	1.11
15. PSQI Global Score	6.26	2.92	1–18	.507**	–.146*	.375**	.354**	.471**	–.002	.011	.008	.189**	.005	.169*	.186**	–.017	–.066	–.165*	6.31	2.96
16. All-nighters (1=yes, past 2 wks) [†]	–	–	–	.252**	–.077	.102	.059	.025	.066	.136	.075	–.053	.083	.135	.157*	.166*	–.104	–.157*	.12	.32
17. Morningness (Owl-Lark score)	49.17	8.74	20–73	–.215**	.089	–.271**	–.154*	.023	–.120	–.202**	–.147*	–.059	–.078	–.095	.046	–.049	.134	.120	49.54	8.80
18. Daytime Sleepiness [§]	.00	11.26	–23.2–40.1	1	–.274**	.379**	.347**	.409**	.066	.104	.097	.135	–.032	.123	.071	–.129	–.058	–.140*	–.13	11.31
19. Cognition Index [§]	.00	2.55	–5.8–6.8	–.237**	1	–.176*	–.187**	–.120	–.217**	–.139*	–.101	–.057	–.003	–.141*	–.104	–.158*	.293**	.310**	.16	2.62
20. Classes Missed [†]	1.64	.73	1.00–4.58	.353**	–.201**	1	.347**	.131	.108	.062	.093	.095	.088	.023	.033	.108	–.204**	–.279**	1.60	.69
21. Health Problems	4.62	3.17	1–23	.287**	–.164**	.282**	1	.244**	–.037	–.048	.021	–.007	–.072	.033	.144*	–.132	–.036	–.145*	4.66	3.29
22. Mood (DASS score)	20.04	16.54	0.00–82.00	.432**	–.136*	.137*	.228**	1	–.103	–.091	–.046	.159*	.092	.065	.093	–.145*	.069	.045	20.12	16.90
23. Alcohol Use (Amount)	5.57	4.10	0–24	.068	–.212**	.131*	–.022	–.069	1	.637**	.566**	–.046	.322**	.167*	.207**	.367**	–.247**	–.351**	5.40	4.12
24. Alcohol Use (Severity)	2.14	2.10	0–12	.124*	–.112	.123	–.008	–.046	.639**	1	.713**	–.002	.298**	.138*	.168*	.176*	–.123	–.098	2.05	1.98
25. Alcohol Use (Frequency)	65.91	58.21	0–365	.093	–.120	.142*	.042	–.031	.561**	.687**	1	.033	.331**	.146*	.195**	.106	–.164*	–.134	62.06	51.77
26. Caffeine Use (Frequency)	145.04	145.91	0–365	.103	–.085	.068	–.007	.175**	–.061	–.002	.069	1	.109	.258**	–.001	–.125	.056	.049	142.12	144.17
27. Tobacco Use (Frequency)	26.73	80.01	0–365	–.003	–.005	.083	–.057	.087	.332**	.303**	.360**	.114	1	.283**	.106	.302**	–.129	–.126	26.66	83.55
28. Energy Drink Use (Frequency)	18.13	54.42	0–365	.125*	–.151*	.085	.058	.078	.129*	.114	.111	.264**	.268**	1	.008	.141*	–.141*	–.222**	19.09	57.87
29. Stimulant Use (Frequency)	17.74	70.77	0–365	.042	–.130*	.019	.157*	.121	.212**	.158*	.131*	.022	.083	.013	1	.008	–.085	–.048	17.39	70.52
30. Sex (1=male) [†]	–	–	–	–.086	–.161*	.139*	–.129*	–.117	.392**	.219**	.156*	–.125*	.309**	.114	.057	1	–.192**	–.249**	.36	.48
31. Cumulative GPA estimate	3.24	.40	2.00–4.00	–.044	.305**	–.206**	.000	.105	–.268**	–.126*	–.185**	.107	–.116	–.109	–.054	–.221**	1	.613**	3.29	.39
32. Semester GPA (N=203)	3.24	.48	1.13–4.00															1	3.24	.48